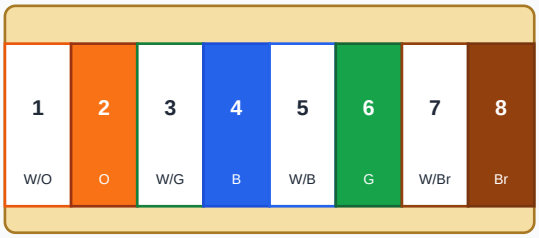
ETHERNET WIRING STANDARDS

Question 1 — RJ-45 pin layout: T568A and T568B

An Ethernet cable with an RJ-45-style 8P8C plug contains eight copper conductors arranged as four twisted pairs. T568A and T568B are two accepted wiring standards. They use the same four pairs but place the green and orange pairs in differe

Pin → color
1: White/Green
2: Green
3: White/Orange
4: Blue
5: White/Blue
6: Orange
7: White/Brown
8: Brown

bers and Ethernet Wire Colors



Pin → color

1: White/Orange
2: Orange
3: White/Green
4: Blue
5: White/Blue
6: Green
7: White/Brown
8: Brown

View the plug consistently: contacts facing you and the latch away from you; left-to-right is pins 1–8.

T568A, pins 1–8: White/Green, Green, White/Orange, Blue, White/Blue, Orange, White/Brown, Brown.

T568B, pins 1–8: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.

For a normal straight-through patch cable, use the same standard at both ends (A–A or B–B). Both standards work for Ethernet when correctly terminated; T568B is common in many installations. Keep the wire pairs twisted as close to the plug as practical.

Question 2 — Ethernet standards comparison

The following table compares common copper Ethernet standards. The maximum channel length shown is the typical Ethernet limit for a compliant twisted-pair link, including the installed cable and patch cords.

Standard	Speed	Cable type	Max. length	Connector
10BASE-T	10 Mb/s (10 Mbps)	Cat3 or better twisted-pair; Cat5e/Cat6 also work	100 m (328 ft)	RJ-45 / 8P8C
100BASE-TX	100 Mb/s (100 Mbps)	Cat5 or better, typically Cat5e/Cat6 twisted-pair	100 m (328 ft)	RJ-45 / 8P8C
1000BASE-T	1 Gb/s (1000 Mbps)	Cat5e or better, commonly Cat5e/Cat6 twisted-pair; all four pairs used	100 m (328 ft)	RJ-45 / 8P8C

How to read the names

In 1000BASE-T, “1000” means 1000 megabits per second, “BASE” means baseband signalling, and “T” indicates twisted-pair copper cable. Ethernet devices usually negotiate the fastest speed supported by both ends and the installed cable. A gigabit router connected to an older 100BASE-TX device will normally use 100 Mb/s on that link.

Note: The maximum length assumes correct cable quality, termination, and installation. Poor terminations, electrical interference, or damaged cable can cause a link to run slower or fail.

Question 3 — Connecting two laptops directly

Traditional answer: Use an Ethernet crossover cable when connecting two laptops directly if their network ports do not support Auto-MDI/MDI-X. A crossover cable swaps the transmit and receive pairs at one end so that one laptop's transmit pair connects to the other laptop's receive pair.

Straight-through versus crossover

Cable type	Wiring	Typical use
Straight-through	Same pin order at both ends (T568A–T568A or T568B–T568B).	Laptop to switch/router; console to switch.
Crossover	T568A at one end and T568B at the other, crossing the relevant transmit/receive pairs.	Older direct connections between similar devices, such as laptop-to-laptop, without auto-sensing.

Modern equipment note: Most modern laptop Ethernet adapters and network devices support Auto-MDI/MDI-X, which automatically adjusts transmit and receive pairs. Because of this, a normal straight-through cable often works for a direct laptop-to-laptop connection today. If you are following a basic networking lab or using older hardware, a crossover cable is the traditional choice.

Step-by-step connection

Step 1: Check whether both laptops have Ethernet ports; if not, use compatible USB-to-Ethernet adapters.

Step 2: For older equipment without Auto-MDI/MDI-X, connect them with a crossover cable. For modern equipment, a straight-through cable will often work.

Step 3: Enable Ethernet on both laptops and make sure the adapters show a connected link.

Step 4: Set compatible IP addresses on the same subnet, or use Internet Connection Sharing if one laptop is intentionally providing a shared connection.

Step 5: Test the connection by pinging the other laptop's IP address and checking that file sharing or the intended service is permitted by the firewall.

Question 4 — Three real-world devices using Ethernet

These are common examples. The exact supported standard depends on the model and its network port, so check the device label or technical specifications when you need a precise answer.

Device	Likely Ethernet standard	Why Ethernet is useful
Home Wi-Fi router / broadband router	1000BASE-T (Gigabit Ethernet) is common on LAN ports in modern routers; some entry-level or older models may offer 100BASE-TX.	A desktop PC, smart TV, or switch can be connected by Ethernet for a stable wired network.
College desktop computer or computer-lab PC	1000BASE-T is common on modern PCs; older systems may support only 100BASE-TX.	Gigabit Ethernet is useful for fast file access, online classes, downloads, and shared lab resources.
Network printer or POS terminal	100BASE-TX or 1000BASE-T, depending on the model; many printers and POS devices use 100BASE-TX or Gigabit Ethernet.	A wired connection can provide dependable printing or payment/inventory communication without relying on Wi-Fi.

Conclusion

T568A and T568B specify the order of the eight wires inside an Ethernet plug. 10BASE-T, 100BASE-TX, and 1000BASE-T support 10 Mb/s, 100 Mb/s, and 1 Gb/s respectively, with a typical maximum copper link length of 100 metres. A crossover cable is the traditional choice for directly connecting similar devices without Auto-MDI/MDI-X, while modern equipment often works with a straight-through cable.